WHITECODE LABS • SYSTEM ARCHITECTURE

WhiteCode Meet Automation

End-to-End Flow & Key Specification: Automated Google Meet Discussion Capture, Multilingual AI Processing & Tasksheet Logging

1 Executive Overview & Problem Solved

Eliminates manual employee timesheet entry by tracking actual Google Meet participation, transcribing **English, Marathi (मराठी), and "Mar-glish"** code-switching verbatim with **Google Gemini 3.6 Flash**, classifying projects, and pushing validated hours directly into individual corporate tasksheets and the WhiteCode HR REST API.

2 Complete Keys & Credentials Reference

Key / Credential	Location	Purpose & How It Is Used in the Backend
Google OAuth Web Client (`client_id`, `client_secret`)	`credentials.json` (Project Root)	Loaded at boot to instantiate `OAuth2Client`. Authenticates the application with Google Calendar & Google Meet APIs.
Google OAuth Tokens (`access_token`, `refresh_token`)	`token.json` (Project Root)	Stores authorized user tokens (scopes: `calendar.readonly` & `meetings.space.readonly`). Automatically persisted upon refresh via `oauth2Client.on("tokens")`.
`GEMINI_API_KEY`	`.env`	Authorizes calls to **Google Gemini 3.6 Flash** via Google AI Files API for: (1) High-fidelity audio transcription in Marathi & English, and (2) Corporate discussion topic extraction.
`PUBLIC_WEBHOOK_URL`	`.env`	Public HTTPS endpoint (Cloudflare/ngrok tunnel) registered with Google Calendar to receive instant push webhook notifications on meeting schedule changes.
`EXTERNAL_TASKSHEET_API_KEY`	`.env`	Bearer token / API key used to push generated tasksheet rows to WhiteCode's live HR REST API (`/api/tasksheet/entries`).
`EXTERNAL_SYNC_ENABLED`	`.env`	Toggle (`true` / `false`) enabling automatic push to external corporate servers.

3 Complete End-to-End System Flow

1 Meeting Discovery

Google Calendar webhook triggers on scheduled events, or Chrome Extension instantly detects impromptu calls on `meet.google.com`.

2 Dual Audio & Captions

Chrome Extension records 128kbps Opus audio & observes DOM captions, stripping Google Meet system toasts ("mic muted", "user left").

3 Multilingual Gemini STT

Audio is uploaded to Gemini Files API. Gemini 3.6 Flash transcribes verbatim Marathi Devanagari script and English technical keywords.

4 Topic Extraction & AI

Gemini analyzes transcript vs. project catalog; outputs Project ID, 2–5 bullet topics discussed, and formal corporate task sentences.

5 Duration Interval Math

Mathematical interval union merges overlapping reconnects/multi-device joins so employees are never overbilled or double-counted.

6 Individual Tasksheet Sync

Creates entry in each attending employee's tasksheet with formatted effort ("1 hr 30 mins") & pushes to WhiteCode HR API.

Overlapping Interval Duration Formula:

Ensures multi-device attendance (phone + laptop) or network disconnects are merged: $$\text{Total Duration} = \sum (\text{Merged Interval End} - \text{Merged Interval Start})$$ Waits until participant leaves call before generating tasksheet entry.

Identity Disambiguation:

3-tier matching: (1) `employee_email_aliases` lookup, (2) Direct email match, (3) Heuristic token scoring (+2000 for distinctive tokens, -1000 for mismatched names like Suraj Dantrao vs Suraj WhiteCode).

4 Backend Processing Architecture (`src/index.ts`)

Google Calendar Webhook & Watcher

Webhook at `/google/calendar/webhook` verifies `x-goog-channel-id` against SQLite table `calendar_watch_channels`. Upon reception, it queries `calendar.events.list`, resolves Google Meet URLs (`meet.google.com/xxx-yyyy-zzz`), and stores meeting records in SQLite with status `DETECTED`.

Multilingual Audio Pipeline (`gemini.service.ts`)

`POST /api/meet/audio-chunk` receives multipart WebM audio from extension. It uploads the buffer to Google AI Files API and invokes Gemini 3.6 Flash. Spoken Marathi is transcribed into Devanagari script, English tech terms (API, database, UI, test) remain in English, and ambient noise is filtered.

AI Discussion Extraction & Translation

When finalized, Gemini classifies the meeting against available projects and translates Marathi discussion points into professional corporate English:
"Attended project meeting to discuss client information support requirements and automated data entry workflow."

Individual Tasksheet & External Push

Validates that duration > 0 and participant has left the call. Creates individual entries in `tasksheet_entries` for each employee. If `EXTERNAL_SYNC_ENABLED=true`, immediately posts to WhiteCode REST API with automatic background retries for failed syncs.

5 Chrome Extension Flow (Manifest V3)

Component	Action & Method	Key Implementation Details
Content Script (`content.js`)	Audio Recording + DOM Captions	Captures tab microphone using `navigator.mediaDevices.getUserMedia` (48kHz, Opus). Observes live captions via `MutationObserver`. Scrapes participants from video tiles & people panel.
Toast Filter (`content.js`)	Regex Noise Suppression	Filters out system messages: *"microphone is on/off"*, *"X left call"*, *"rate audio/video"*, *"you are muted"* to prevent corrupting corporate discussion logs.
Popup UI (`popup.html` / `popup.js`)	User Controls & Live Dashboard	Shows backend connection status, active meeting code pill, live scrolling transcript box, and the **"Finalize & Fill Tasksheet"** trigger button.

6 Core REST Endpoints & Database Schema

Method	Endpoint	Payload / Params	Action
GET	`/oauth2callback`	`code`, `scope`	Persists refreshed Google tokens into `token.json`.
POST	`/api/meet/audio-chunk`	`audio` (WebM), `meetingCode`	Transcribes audio with Gemini and auto-updates tasksheets.
POST	`/api/meet/finalize-discussion`	`meetingCode`, `participants`	Extracts 2–5 discussion topics and creates tasksheet entries.
GET	`/tasksheets`	`employeeId`, `date`	Visual dashboard to inspect, filter, and edit employee timesheets.
POST	`/api/sync/external`	None	Triggers 2-way sync with WhiteCode's HR/Tasksheet REST API.

SQLite Schema (`meetings.db`): `meetings` (records, status, project), `meeting_transcripts` (verbatim chunks), `meeting_participants` & `meeting_participant_sessions` (intervals), `employees` & `employee_email_aliases` (directory), `projects` (catalog), `tasksheet_entries` (hours, topics, sync_status).

7 Quick Start in 4 Steps

1. **Dependencies & Keys:** Place `credentials.json` in root, set `GEMINI_API_KEY` in `.env`, run `npm install`.
2. **Start Backend:** Run `npm run start` (server starts on port 3000). Visit `http://localhost:3000/` to grant Google permissions.
3. **Load Extension:** Open `chrome://extensions`, enable Developer Mode, click "Load unpacked", select `chrome-extension` folder.
4. **Live Test:** Join any Google Meet call, click extension icon, click "Start Discussion Tracking", speak, click "Finalize & Fill Tasksheet", and view updated entries on `/tasksheets`!